C++ LAB SUBMISSION REPORT

Data Structures & OOP Implementation

Task 1: List Implementation and Length Calculation

Implementation of a fixed-size array-based list to input numbers and display the list along with its length.

PROGRAM CODE

```
#include <iostream>
using namespace std;

class List {
private:
    int arr[10];
    int size;
    int n;
public:
    List() {
        size = 0;
    }
    void input() {
        cout << "Enter size: ";
        cin >> n;
        size = n;
        cout << "Enter " << size << " numbers: ";
        for (int i = 0; i < size; i++) {
            cin >> arr[i];
        }
    }

    void display() {
        cout << "List: ";
        for (int i = 0; i < size; i++) {
            cout << arr[i] << " ";
        }
        cout << endl;
    }

    int Length() {
        return size;
    }
};

int main() {
    List list;
    list.input();
    list.display();

    cout << "Length of list is: " << list.Length() << endl;

    return 0;
}
```

OUTPUT

```
Enter size: 5
Enter 5 numbers: 10 20 30 40 50
List: 10 20 30 40 50
Length of list is: 5
```

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Task 2: Delete Element by Index

Extends the List class to remove an element at a specified index position and shift subsequent elements left.

PROGRAM CODE

```
#include <iostream>
using namespace std;

class List {
private:
    int arr[10];
    int size;
    int n;
public:
    List() {
        size = 0;
    }
    void input() {
        cout << "Enter size: ";
        cin >> n;
        size = n;
        cout << "Enter " << size << " numbers: ";
        for (int i = 0; i < size; i++) {
            cin >> arr[i];
        }
    }
    void remove() {
        int position;
        cout << "Enter index to delete: ";
        cin >> position;

        for (int i = position; i < size - 1; i++) {
            arr[i] = arr[i + 1];
        }
        size--;
    }
    void display() {
        cout << "List: ";
        for (int i = 0; i < size; i++) {
            cout << arr[i] << " ";
        }
        cout << endl;
    }

    int Length() {
        return size;
    }
};

int main() {
    List list;
    list.input();
    list.display();
    list.remove();
    list.display();
    cout << "Length of list is: " << list.Length() << endl;

    return 0;
}
```

OUTPUT

```
Enter size: 5
Enter 5 numbers: 12 24 36 48 60
```

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Task 3: Delete Element by Value

Searches for a specific value within the list, identifies its position, and removes it from the array.

PROGRAM CODE

```
#include <iostream>
using namespace std;

class List {
private:
    int arr[10];
    int size;
public:
    List() {
        size = 0;
    }
    void input() {
        cout << "Enter size: ";
        cin >> size;
        cout << "Enter " << size << " numbers: ";
        for (int i = 0; i < size; i++) {
            cin >> arr[i];
        }
    }
    void remove() {
        int value;
        cout << "Enter number to delete: ";
        cin >> value;
        int position = -1;
        for (int i = 0; i < size; i++) {
            if (arr[i] == value) {
                position = i;
                break;
            }
        }
        for (int i = position; i < size - 1; i++) {
            arr[i] = arr[i + 1];
        }
        size--;
    }
    void display() {
        cout << "List: ";
        for (int i = 0; i < size; i++) {
            cout << arr[i] << " ";
        }
        cout << endl;
    }
    int Length() {
        return size;
    }
};

int main() {
    List list;
    list.input();
    list.display();
    list.remove();
    list.display();
    cout << "Length of list is: " << list.Length() << endl;
    return 0;
}
```

OUTPUT